

ACTIVITY - 1

1. Medical Image Enhancement :-

A doctor receive an x-ray image in which the bones are visible, but the darkest tissues have poor visibility.

1. I would recommend power-law (gamma) Transformation ($\gamma < 1$).

2. It is suitable because it enhances the dark region of x-ray while preserving brighter bone details.

3. The darker tissues become clearer, contrast improve and hidden details become more visible.

2) Night - Time Surveillance:

A CCTV camera captures a parking lot lot of night. Most pixel values are concentrated between 0 and 40 making the images very dark.

1. Transformation - Log Transformation

2. Log transformation expands low-intensity (dark) pixel value, making dark areas brighter. Negative Transformation only inverts brightness and does not improve visibility.

3. The parking lot appears brighter, objects become more visible and details in dark region are better and easier to identify.

3. Fingerprint Recognition:

A finger image contains many fine ridge details but the contrast between the ridges and valley is very low.

1. Contrast Stretchings

2. It increase the contrast between fingerprint ridges and valley, making the details more distinguishable.

3. Log transformation would brighten the overall image but would not significant improve the contrast between ridges & valley making finger print

